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Automated Prediction of Liver Disease using Machine Learning (ML) Algorithms

Aviral Srivastava, V Vineeth Kumar, Mahesh T R, V. Vivek
Computer Science and Engineering, JAIN (Deemed-to-beUniversity), Bengaluru, India
15bt6cs016@jainuniversity.ac.in, 18btrcs031@jainuniversity.ac.in, t.mahesh@jainuniversity.ac.in, v.vullikanti@jainuniversity.ac.in

Abstract*—Information systems and strategic tools have lately been offered as new strategies in medical research to improve the disease detection process. Enzyme activation, bile synthesis, lipid metabolism, and vitamin, glycogen, and mineral storage are all crucial functions of the liver. Because liver problems are difficult to diagnose in the early stages due to a lack of particular symptoms, they are commonly neglected. Hyperbilirubinemia is common symptom in many liver diseases, and it's tough to tell the difference early on. However, this is not guaranteed, and understanding enzyme levels is necessary to detect and confirm the presence of liver illness. A variety of machine learning (ML) algorithms are being used to predict liver diseases. We recommend employing Logistics Regression (LR), Naive Bayes Model (NB), K-Nearest Neighbor (Knn) in this project. The data was separated into two categories: patients with liver disease and sicknesses the most accurate machine learning method was used to predict the final result. The different algorithms are compared against various performance metrics and the best one is identified.

Keyword: Liver disease, Naive Bayes, logistic regression, Nearest Neighbor, Accuracy

I. INTRODUCTION

Liver is the second most important gland of our body. Its located in the right part of the abonnement approximately 1.5 kg and has a width of about 15 cm. The entire blood leaving in the stomach and intestine pass via liver. Liver helps in the digestive system so liver is also called body's chemical "factory". Liver infections are so many types such as Hepatitis A, Hepatitis B and Hepatitis C. Liver disease is the dangerous disease in the world Liver disease can be inherited liver disease are cause by alcohol, obesity, diabetes and also impurewater [1].

It shows many symptoms such as vomiting, fatigue, abdominal pain nausea and swelling. Live rproduce bilirubin chemical due to increase in bilirubin, jaundice will occur [2]. For enhanced the health analysis through data mining approach is being to classify the liver disease [3].By data mining with machine learning the doctors and medical provides to extract the correct information to suggest for effective treatment [4].

Liver infections have created a major information, for example, electronic wellbeing records, metabolomics analyses as well as report that includes patient clinical data, what's more problems [5]. Nonetheless, these information should be examined and incorporated assuming that they are going to create models regarding physiological instruments of pathogenesis [6]. In this study, we make use of AI dependent classifier for enormous databases in the area of liverto Predict what's more remedial revelation, logistic regression, k nearest neighbor and NB, information mining methods are utilized with the new model for expectation of liver sicknesses [7]. The presentation of these mentioned classifier strategies are assessed with accuracy, recall and precision.

II. RELATED WORK

Liver illnesses likewise come in the classification of constant infections. An enormous number of contaminations influence the liver which prompted the different liver illnesses [8]. The passing s because of the liver sicknesses has touched 208185 or 8.34% of complete demise in Egypt. The organ liver is probably the most grounded organ in the body that is located on the right half of the midsection [9].The insufficiency of Insulin leads to high blood glucose levels and slowed carbohydrate, lipid, and protein metabolism [10].The liver along with these organs collaborate to assimilate as well as to process food. Liver cleans the blood starting from the stomach related lot. Similarly, cleans synthetics and uses drugs. The organ liver covers bile that ends up again in inward parts [11]. Also, the liver also makes proteins basic for blood coagulating and is also responsible for the clearing of microorganisms and exogenous antigens from the fundamental stream. Liver contamination is suggested as hepatic infirmity.

Liver contamination is an overall terminology that covers every one of the standard issues that achieve the liver to disregard to play out its allocated limits. This concentrate basically examines around five kinds of liver illnesses like alcoholic liver harm (ALD), liver cirrhosis (LC), essential hepatoma (PM), HCC and cholelithiasis (C) [12].

Type-1 diabetes is a kind of diabetes that affects about 5% of diabetics. Symptoms can appear out of nowhere and get worse over time. Frequent urination, increased thirst, bedwetting (in youngsters), weight loss, weakness, extreme hunger, irritability, fatigue and also, blurred eyesight are among symptoms of Type 1 Diabetes. Type2 disease can be managed with a combination of diet and exercise, as well as a medicine that improves sensitivity insulin production of the body. However, insulin production that is natural may be insufficient at times, necessitating insulin injections to maintain normal blood glucose levels. If you are old i.e; 45 years of age or even older, if you have diabetes history of the family, or if you are overweight or even obese, then definitely you should consult a doctor. One of the primary driver of expanded liver illnesses in Egypt [3] is weight, breathe in of unsafe gases, admission of debased food, unreasonable utilization pickles and medications, liquor [13] .The target of this article is to suggest a Machine learning strategies in light of Classification of Liver issues.

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There are three algorithms being implemented and evaluated with respect to various performance metrics and the best model is recommended.

III. METHODOLOGY

Our aim for this research consists of

- Reduction of number of deaths and more accurate diagnosis
- Expanded comfort for anticipating a liver infection
- More exact conclusion of liver infection by the specialists
- Improvised algorithm or medical tool for fast Liver infection detection at early stage
- Machine learning strategies in light of Classification of Liver issues for diminish trouble on specialists

The goal of article is to give a performance comparison of machine learning classification algorithms for liver illness prediction using datasets taken from website Kaggle. It comprised of ten input variables, one output variable and 116 records of people. Following the selection of dataset as shown in figure 1, each dataset is subjected to a preprocessing phase that serves two purposes: restoring missing values, transforming the class feature from numbers to categorical, as some algorithms don't work with numerical class features. Then, when machine learning (ML) methods are being applied to the dataset, the results are evaluated using various performance metrics to show which techniques perform better. F1 score, precision, and accuracy are being used to evaluate performance of ML algorithms on liver datasets.

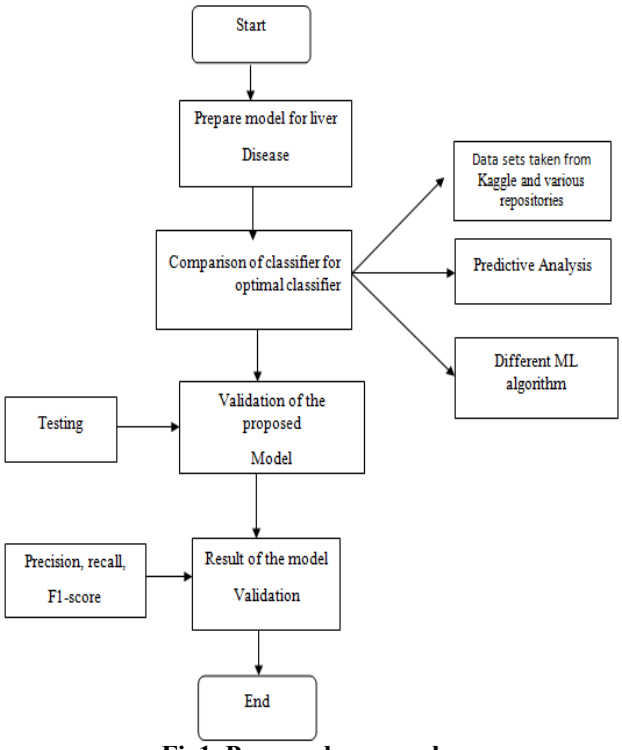

Fig1. Proposed approach

The following various ML models have been successfully implemented.

A. K nearest-neighbor (KNN)

For both classification and regression, kNN is a supervised learner. Classification is a categorical target variable prediction job. Models of classification learn to classify any new observation. This assigned class can only be correct or incorrect, not both. The iris dataset is a typical example of classification, in which physical characteristics of plants are used to predict their species. Regression is a prediction job with a numeric target variable. The Kaggle Housing Prices Challenge is a well-known example of regression. Participants in this machine learning competition attempt to forecast house sales prices based on a variety of independent variables.

Model complexity should be considered as a last criterion for describing machine learning models. Machine learning, particularly artificial intelligence, is booming right now, and it's being used for a variety of complex jobs including analyzing text, photos, and speech, as well as self-driving automobiles.

Advanced and complicated models, such as neural networks, are likely to be able to learn more than a k-Nearest Neighbors model. Those sophisticated models, after all, are excellent learners. However, keeping in mind that this level of sophistication comes at a cost. You'll need to spend a lot more effort developing the models to make them fit your predictions.

The supervised ML algorithm k-nearest neighbor is used to address classification and also, regression problems. It is, however, mostly employed to solve categorization difficulties. KNN is a non-parametric, slow learning method.

B. Naive Bayes (NB)

The Bayes Theorem, developed by Reverend Thomas Bayes in the 1760s, is the foundation of Naive Bayes. Its popularity has soared in the recent decade, and the algorithm is now frequently employed in academia, government, and business to solve challenges. In practical machine learning, a Naive Bayes classifier combines a number of desired qualities. We'll go into the reasoning for this a little later. Let's get started by learning how a Naive Bayes algorithm works and then using the scikit-learn module to create it in Python.

The supervised machine learning methods that Naive Bayes belongs to are largely utilized for classification. "Supervised" means that the algorithm is trained with both input features and categorical outputs in this case.

C. Logistic regression (LR)

The regression estimations are used to illustrate how one dependent variable interacts with one or more independent variables.

The objectives of regression are: (1) Possible to suggest an outcome (dependent) variable using a set of predictor variables? (2) Which variables in particular are significant predictors of the outcome variable, and how they are really going to influence the output variable..

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IV. RESULTS AND DISCUSSION

The chosen data set has been split into 2 parts: the first one is training data which contains 80% of the total data set, the second one is the testing data which contains 20% of overall data set. Once the data is prepared, the methods are applied and later confusion matrix has been discovered. The outcomes were compared with respect to accuracy of different algorithms. Confusion matrix values determines the accuracy It is shown in figure2.

	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

Fig 2. Confusion Matrix

Confusion matrices were created by making use of model's predictions and the values are described in table1.

Table I. Confusion matrix values

ML	TPs	FPs	TNs	FNs
Logistic Regression	81	3	6	26
Naive Bayes	31	53	30	2
Knn	75	9	7	25

The confusion matrix of the following three algorithms namely, LR, NB, and knn and have been shown in fig 3.

All the three algorithms are executed against different performance metrics. The performance percentage of these is shown in figure 4.Every machine learning pipeline has performance measurements. They tell you if you're progressing and give you a number. A metric is required for all machine learning models, whether linear regression or a SOTA approach like BERT.

Any project should include an evaluation of your machine learning method. Any model may produce a satisfactory result when examined using a metrics accuracy score, however it may produce unsatisfactory outcomes when examined using other metrics. We'll go over the many sorts of evaluation metrics available in this article.

In classification when we say "accuracy," we're usually referring to precision.

$$Accuracy = \frac{No\ of\ Correct\ Predictions}{Total\ No\ of\ predictions} \quad (1)$$

The precision is given by

$$Precesion = \frac{TruePositives}{TruePositives+FalsePositives} \quad (2)$$

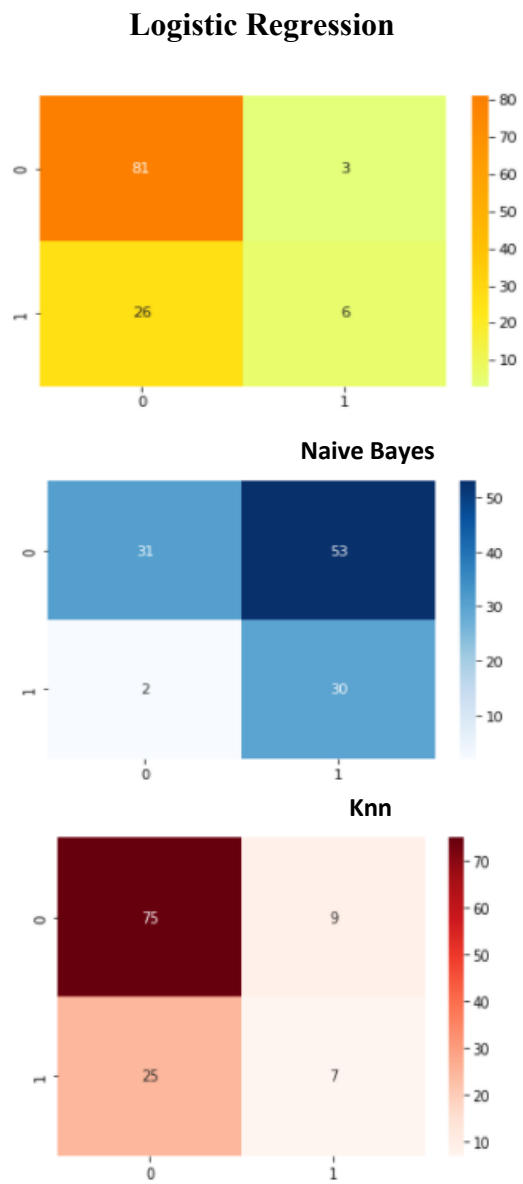

Fig 3. Confusion Matrix

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In the presence of a class imbalance, precision as well as recall both provide a superior approach of measuring model performance. They'll inform us that the model isn't really useful in our situation. The recall is given by

$$Recall = \frac{TruePositives}{TruePositives+FalseNgatives} \quad (3)$$

The Harmonic Mean which takes care of both precision as well as recall into consideration. It is stated as follows:

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$$F1 = 2 * \frac{1}{\frac{1}{precision} + \frac{1}{recall}} \quad (4)$$

Among the stated three ML algorithms. after evaluated against accuracy as shown in figure 5, it is very much noticed that LR has provided comparatively better accuracy as compared with the remaining algorithms.

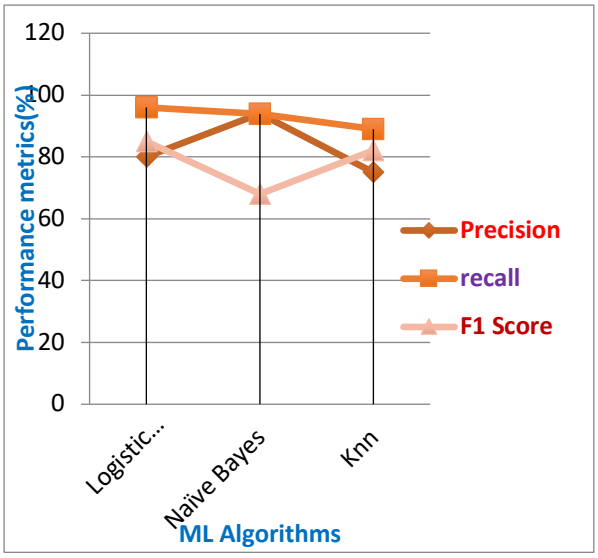

Fig 4. Performance metrics

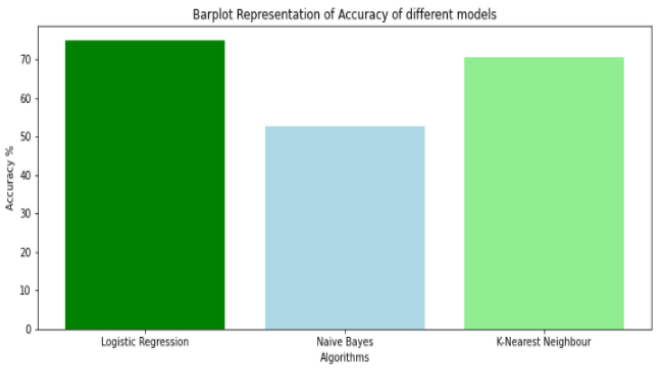

Fig 5. Accuracy of ML algorithms

V. CONCLUSION

In this work, a review of different ML algorithms for predicting liver disease in patients is done. The three ML techniques used includes Naive Bayes, KNN and Logistic Regression. The system was implemented using all the models as well as their performance has been evaluated. The accuracy of all these different algorithms have been compared against different performance metrics. Logistic Regression model resulted in the maximum accuracy of 75% when compared against the other various algorithms.

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